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Renewable Energy Based Smart Irrigation System

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Abstract

Agriculture is the primary occupation in India and is called India's backbone. But of late, a lot of problems are being faced in agriculture by the farmers. One of the major problems being water scarcity. As per surveys, almost 20 percent of the agricultural land is wasted due to water scarcity and becomes a barren land. Thus, this research gives an idea of smart irrigation system. This irrigation system uses three sensors namely temperature sensor, humidity sensor and soil moisture sensor, and fuzzy logic is used to operate the solenoid valve. The data from the sensors is then sent to the cloud by using adafruit.io and the farmer can view the moisture level, humidity level and temperature recorded by these sensors. All the operations are governed by an Arduino and the power supply for the Arduino is given by a solar panel which uses LDR (Light Dependent Resistor) and makes it into an automatic tracking system. The crops are grown in two tubs and comparison of growth of plant with automated irrigation and normal irrigation is carried out.

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1. Introduction

Agriculture is the major backbone of India as it satisfies the need of food for the people. Though many agricultural lands are being destroyed, there are still many who are not willing to sacrifice agriculture as they provide basic requirement for the people. Agriculture is facing many problems as of today and many researches are being carried out in order to improve the agricultural practices. Some researchers have made a Zigbee module to control three sensors to give right amount of water to the plant which also incorporated by transfer of feedback to the user through the mobile app and some have used only a soil moisture sensor for automatic drip irrigation. Then a three level moisture sensor which detects moisture at three different depths and accordingly the water is allowed to the crop. The soil pH was also read using image processing. An attempt to detect water requirement by moisture sensor and used solar panel to pump the motors. Then later fuzzy logic was used which had three sensors (soil moisture, humidity and temperature sensors) and the solar panels are connect to run the pumps.

The automated irrigation is done by using fuzzy logic and the basic controller of all is Arduino as it controls the sensors and the solenoid valve. As per the survey taken, we came to know that the voltage from the solar panel is insufficient to power the pump and it is expensive to buy batteries to store power from solar panel. Thus, we use the solar panel to power the Arduino. The solar panel is an auto tracking system which is made by LDR (Light Dependent Resistor). The plants are grown in both tubs and separation is shown in order give a view on two different lands and the plants are grown using automated irrigation system in one and the normal irrigation on the other. The quality of crops and the amount of water used to grow are compared.

The paper is about Renewable Energy based Irrigation System to avoid disadvantages and limitations of present irrigation systems where the main problem is disproportionate distribution of water to crops, hence consumption of excess electricity. This proposed System is designed to increase the efficiency of water and power by solar panels to make it eco-friendly. In addition, this can be implemented on large or small scales.

Here, a distributed network of sensors is used to detect the moisture content of the soil. These sensors are connected to a control unit which is responsible for controlling as well as monitoring the whole irrigation process. Depending on weather conditions a decision based on fuzzy logic will be made regarding the need to irrigate the soil.

Of late, trends in consumption of more electricity and water were witnessed. Irrigation systems developed in the past were either high budgetary or followed traditional, non-renewable sources of electricity. Moreover, Irrigating farms requires high intervention of human, whereas in the proposed system human intervention is minimal.

2. Related Work

Joaquin Gutierrez proposed an irrigation system ^[17] that uses photovoltaic solar panel to power system because electric power supply would be expensive. For consumption of water in an effective and optimal way, an algorithm was developed with threshold value of temperature and soil moisture programmed into a micro controller gateway. G.Parameswaran proposed a system ^[9] that helps the farmers to irrigate the farmland in an efficient manner with automated irrigation system based on soil humidity. Humidity sensor is used to find the soil humidity and based on this microcontroller drives the solenoid valve. Irrigation status is updated to the server or localhost using Personal Computer. V.Akubattin designed a system ^[10] to monitor and control the soil moisture and temperature inside a greenhouse. The system controlled by Raspberry Pi detects the soil moisture content and according to which it takes a decision watering the plant or switching the fans of green house. D.Zhang designed a system ^[13] based on data transmission unit (DTU), wireless radio frequency (RF) module and microcontroller, the application of RF module in the acquisition terminal improves the expandability, data of soil moisture are collected by relay station, and then transmitted to the monitoring center by the DTU using GPRS network. A.Kumar presents a smart system ^[14] that uses soil moisture sensor to control water supply in water deficient areas. The sensor, which works on the principle of moisture dependent resistance change between two points in the soil, is fabricated using affordable materials and methods.

3. System Design

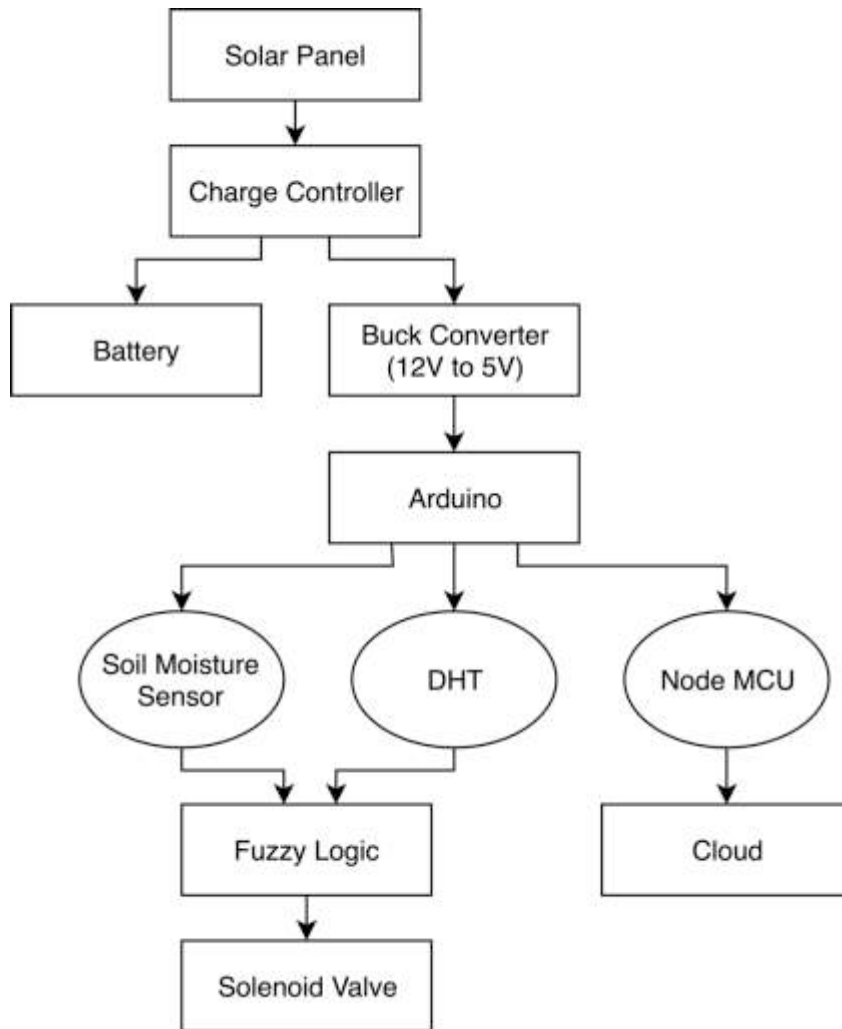


Fig. 1. Renewable Energy Based Smart Irrigation System

4. Methodology

4.1. Solar Panel Assembly

The automatic solar tracking system using the LDR was made in this assembly. The solar panels are connected to Arduino which act as a power source for sensors and the solenoid valve.

4.2. Arduino Assembly

The Arduino is connected to the solar panel which acts as the power source and it is connected to the three sensors namely soil moisture, temperature and humidity sensor. The Fuzzy logic code is written and uploaded into the Arduino. This Arduino is connected to the solenoid valve. The input values are taken from the three sensors and output is given to the solenoid valve. Solenoid valve works on the principle of electromagnetic induction. When the current is applied the solenoid, valve allows the valve to open and remains closed at remaining time.

4.3. Fuzzy Logic

The humidity, temperature and soil moisture sensors are the input values. Four functions of temperature namely “Very Hot”, “Hot”, “Cold” and “Very cold” are taken. For soil moisture and humidity sensors three functions are taken namely “High”, “Medium” and “Low”. Fuzzy logic uses “if” and “else” condition for operating the solenoid valve. There are total of 36 combinations possible from the above three sensors. Based on these conditions, the output value is either “True” or “False”. “True” indicates the current to the solenoid valve and thus water is given to the crops and “False” indicates no current to the solenoid valve and thus the valve remains closed.

4.4. Crop Planting

The crops are planted in two tubs and are grown. The crop in one tub is grown using automated irrigation system and the other one is grown using normal irrigation. The amount of water given to the crops are noted and the quality of the crop are compared after growing. The growing time of plant is one week. A single tube is separated into two regions in order to have two fields of land. So, we use two solenoid valve and the power is given to the respective solenoid valve using the fuzzy logic.

4.5. Cloud Module

This is an Arduino like hardware IO and can be used as an event driven API for networking applications. It is used for Wi-Fi networking (can be used as access point and/or station, host a webserver) and connects to internet to fetch or upload data. This is an excellent system on board for Internet of Things (IoT) applications and can be used for sending sensor data of soil moisture, DHT (Digital Humidity and Temperature) Sensor and was able to run analytics on the data. Adafruit IO is used as an API Provider for uploading the data. It is a cloud service to display our data in real-time, online and make our implementation of this research internet connected.

5. Implementation

5.1. Light Sensing Solar Tracker System Design

The Smart Irrigation System consists of the Light Sensing Solar Tracker System which is at the core of the energy source for the entire system.

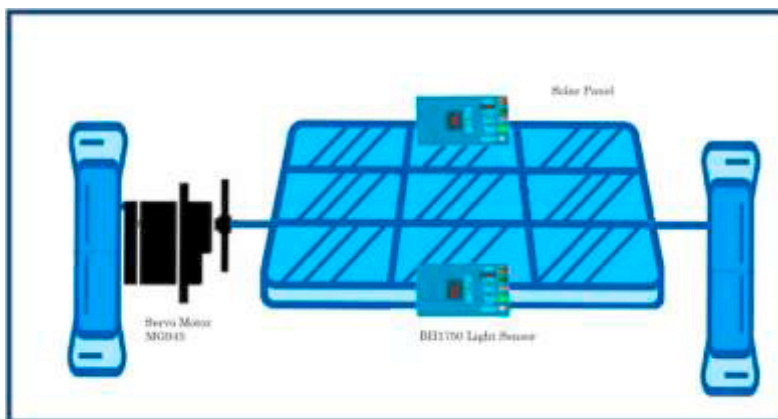


Fig. 2. The Solar Light Tracking System

This system consists of the Solar Panel, A Servo Motor, Two Light Sensors (BH1750 Light Sensor) and an Arduino. The servo motor and the light sensors are connected to the Arduino. The solar panel is attached to the servo motor on

one side and a hinge on the other. This allows for the free rotation of the panel by the servo motor. The motor is designed to rotate the solar panel in either clockwise or anti-clockwise with a total range of 0 – 180 degrees. The allowed rotation of the panel as given by the Arduino to the servo motor is between 20-120 degrees to avoid any zero error and prevent the wire connections to the panel from getting disconnected. The incidence of rays from the sun on the panel is also between these angles for maximum efficiency of the panel and hence the limiting of the angle of rotation is not a concern for energy output from the panel.

There are two BH1750 Light sensors attached to the opposite ends of the solar panel. These sensors relay the light intensity readings on its side to the Arduino. Arduino compares both the intensity values and sends signal to the servomotor to rotate the solar panel to the side (sensor side) which has the higher light intensity reading. The Arduino checks for the intensity value at each angle of rotation. The motor is signalled to stop when sensor values are equal or when it has reached the limiting range of the motor (20 or 120 degree). This is done continuously so that the solar panel can be positioned at maximum energy efficiency incidence angle against the sun rays as the sun rises and moves to set throughout the day. The solar energy generated is stored in a 10000 mAH Battery. This battery gets charged and powers the other components such as Pump, Solenoids, Arduinos 24x7.

Table 1. Power Consumption of Hardware Components.

Component Name	Hours Worked (H)	Rating (mA)	Total mAH
Arduino 1	12	50	600
Arduino 2	12	50	600
Pump	3 (max time)	1500	4500
Solenoid	12 (max time)	100	1200
NodeMCU (WiFi Module)	12	60	720
Total			7620

Table 1 gives us the power consumption of all the hardware components. This total power required is only during the night when the system is to be powered by the battery. During the daytime, enough power is available through the solar panels to charge the battery and power the components and hence the power values for the servo motor, BH1750 sensors are not considered here.

The power requirements for the DHT as well as the soil moisture sensor are taken care of by the Arduino's power requirements and have not been included here. The battery backup available is 10000 mAH. This is more than sufficient to sustain the system throughout the night keeping the plants healthy. As the sun rises, the battery gets charged through the 12V 2A solar panel.

The current output from the solar panel is 2A which fully charges the battery in $8000/2000 = 4$ Hours. Since the solar panel powers the other components also during day time the battery charging time is expected to be impacted with estimated 5-6 hours of charging for full capacity.

5.2. Fuzzy Logic

The renewable based irrigation system uses a fuzzy logic based control where we get the actual sensor readings, In this case the humidity, temperature, and soil moisture sensor values were taken as inputs and were created with three linguistic terms for temperature, three for humidity, and three for soil moisture. The terms were named "High", "Normal", and "Low" for the humidity, soil moisture sensors and the temperature sensor.

5.2.1. Architecture

- Rule Base: Fuzzy logic works around a set of predefined rules given by the experts, to provide the fuzzy system some criteria on which the decision has to be taken. These rules are defined on the basis of linguistic information in reference to our application. Recent findings in the area of fuzzy logic have given

birth to new ways of developing the system. These developments have reduced the number of rules required for the fuzzy system.

- **Fuzzification:** The inputs given to a fuzzy system (in this case the inputs are temperature, moisture content, etc) are called crisp inputs, which are mapped to the fuzzy sets which define the different intervals to which the inputs can belong.
- **Inference Engine:** This is the function which takes in the input and based on the rules provided to the fuzzy system, it gives the decision or gives out the output value for the appropriate action to be taken.
- **Defuzzification:** The obtained output in terms of fuzzy sets, which need to be converted to a crisp value. This process can be handled using various strategies and in order to address any error, proper functions have been used to scale back the error.

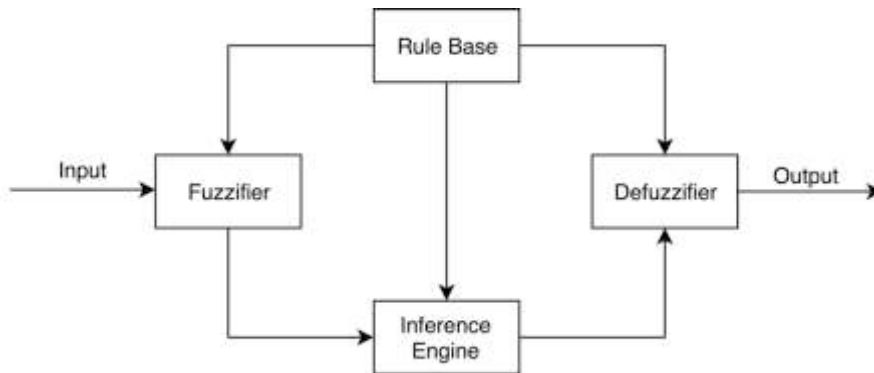


Fig. 3. Fuzzy Logic Architecture

5.2.2. Membership Function

A fuzzy set is converted into a graph which is used to map the membership value between 0 and 1. All possible inputs are represented using a universal set(u), this space is customized for every application.

There are many types of fuzzification functions, the most common ones out of these are:

- Singleton fuzzifier
- Gaussian fuzzifier
- Triangular or Trapezoidal fuzzifier

For our renewable based irrigation system, we used a triangular fuzzifier as a membership function. Triangular function is defined by a lower limit ‘a’, an upper limit ‘b’, and a value ‘m’, where $a < m < b$.

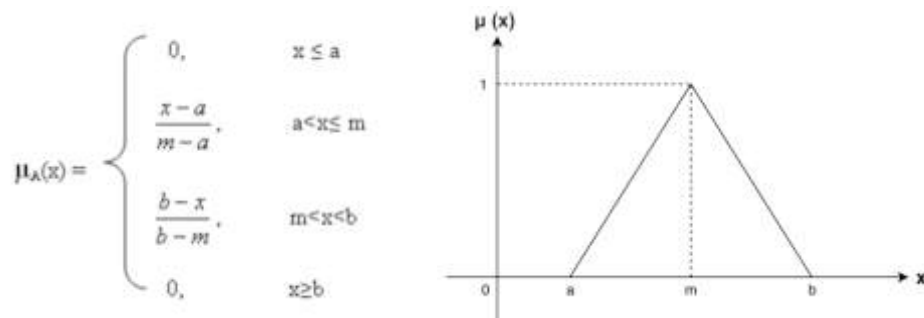


Fig. 4. Membership Function

5.2.3. Fuzzy sets that were used

- Creating fuzzification of soil
 FuzzySet* low_soil = new FuzzySet(0, 50, 50, 75);
 FuzzySet* normal_soil = new FuzzySet(50, 75, 75, 80);
 FuzzySet* high_soil = new FuzzySet(80, 90, 90, 100);
- Creating fuzzification of Humidity
 FuzzySet* low_hum = new FuzzySet(0, 0, 0, 380);
 FuzzySet* normal_hum = new FuzzySet(237, 380, 380, 712);
 FuzzySet* high_hum = new FuzzySet(570, 712, 950, 950);
- Fuzzy output watering duration for plants
 FuzzySet* low_dur = new FuzzySet(0, 0, 0, 0);
 FuzzySet* normal_dur = new FuzzySet(3, 5, 5, 7);
 FuzzySet* high_dur = new FuzzySet(0, 0, 0, 0);

5.3. Internet of Things Integration

Adafruit IO header files contains many APIs such as to connect to a dashboard ^[18]. A dashboard can contain up to maximum of 8 feeds. We have three feeds namely Soil Moisture percentage, Room Temperature, Humidity percentage. Moreover, Adafruit IO logs the values and server responses in the cloud so that even troubleshooting is made effective.

6. Results



Fig. 5. Hardware Setup of the Smart Irrigation System

Table 2. Water Consumption

Days	Normal Irrigation (ml)	Smart Irrigation System (ml)
1 to 3	2400	1550
4 to 6	2400	1225
7 to 9	2400	1140
9 to 12	2400	1175
12 to 15	2400	1115
Total	12000	6205

Table 2 gives us the quantity of water used by the normal method and the smart irrigation method for every 3 day period from day 1 till day 15. The total quantity of water used by the smart irrigation system is 51.7% of the water used by the normal method. So, the quantity of water consumed by this smart irrigation system is 48.3% lesser than the normal method, which is significant in the long term.



Fig. 6. Dashboard – Data Stored and Viewed in the Cloud



Fig. 7. Crops grown using (a) Normal Irrigation; (b) Smart Irrigation System

7. Limitations

The proposed system cannot predict time to harvest, animal intrusion in the field and nutrient level of the plants. Also, the fuzzy logic was designed for crops which can be grown in summer and optimal water is given for those crops. The water needed for the other crops could be different, which would hence require a mild change in the fuzzy logic values.

8. Conclusion

An automated smart irrigation system was successfully developed that can use fuzzy logic and various sensors like soil moisture and DHT Sensor to analyze conditions of the soil and decide whether it should irrigate the farm or not. We also deployed the system without the need of electricity as we used solar energy to power all the Arduino and the sensors. The design of this system addresses the concerns in traditional irrigation systems, for which a considerable waste of resources is noticed. The need for excessive manpower was sorted out via introducing an automated feature for checking the status of the soil, water pumping needs etc. The proposed system is cost-effective and could be easily installed in gardens, roofs, as well large areas for planting crops. Depending on solar energy, as the main source of energy, was an added advantage, as the current trend is to rely more and more on renewable energy sources. Using the Arduino microcontroller added more versatility to the proposed design, as it offers more extendibility and allows the user to add more sensors, with a minor added programming effort. This design can in future be scaled up to suit actual farm sizes, and support the operation without requiring human intervention and man power for irrigation, and can also be used in optimizing the cost of the accessories used in implementing and maintaining the irrigation system.

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